



Continuity, Convexity, Centrality: Axiomatizing and Comparing Location Indices

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Big Picture

- ▶ Ultimately interested in the Fourfold Pattern of Risk Attitudes (FFP):
 - ▶ Risk-seeking on low chance gains, high chance losses; risk-averse otherwise
- ▶ Several related empirical puzzles:
 - ▶ Why does the FFP arises?
 - ▶ Why is it related to evaluation tasks only? (Harbaugh et al., 2010)
 - ▶ Why does it arise in riskless environments? (Oprea, 2024)
- ▶ Prospect theory can only explain the first point
- ▶ My solution: modelling CEs as location indices
 - ▶ Location index: single value best representing numerical data
 - ▶ Mean, median, quantiles, minimum, maximum...
 - ▶ Measures the position of a distribution on the real line

Certainty Equivalents as Location Indices

- ▶ The relation between means and certainty equivalents has deep roots:
 - ▶ Huygens (1657) set the fair price of a lottery equal to its expected value
 - ▶ Bernoulli (1738) moved beyond nominal values (“moral expectation”)
 - ▶ Laplace (1820) linked moral expectations and individual preferences
 - ▶ von Neumann and Morgenstern (1944) acknowledged that an axiomatic treatment of utility is fundamentally a choice regarding the appropriate “mean”
- ▶ Duality between Kolmogorov-Nagumo (1930) and vNM axioms:
 - ▶ de Finetti and Emanuelli (1967) noticed the equivalence between axioms on quasi-linear means by Kolmogorov (1930) - Nagumo (1930) and EU axioms
 - ▶ Direct mathematical mapping between the Associativity Axiom in the theory of means and the Independence Condition in decision theory
- ▶ Since then, the relation has been further generalized:
 - ▶ Hong (1983) moved beyond quasi-linear means to address income inequality and decision-theoretic anomalies like the Allais Paradox
 - ▶ Fishburn (1986) formally unified certainty equivalence and mean values, providing a foundation for non-EU representations

What I do

1. Axiomatic foundation of location indices obtained by minimizing continuous and (strictly) convex loss functions
 - ▶ Statistics: OLS, WLS, maximum log-likelihood principle...
 - ▶ ML: Hedge algorithm, OGD, FTRL, SVM, convex surrogates...
2. Analysis of geometrical properties of the loss function
 - ▶ Asymmetry: uneven penalization of over- vs under-prediction
 - ▶ Convexity: sensitivity to outliers (Hampel, 1974)
3. Behavioral interpretation of such features
 - ▶ Origin asymmetry: first order risk preferences (Segal and Spivak, 1997)
 - ▶ Convexity: centrality preferences
4. FFP as interaction between risk and centrality preferences

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2. Analysis of geometrical properties of the loss function ✗
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 - ▶ Convexity: centrality preferences
4. FFP as interaction between risk and centrality preferences ✗

Setting (1)

- ▶ $D = \{x_1, \dots, x_n; p_1, \dots, p_n\}$ is a *dataset*
 - ▶ $x_i \in \mathbb{R}$ are *observations* s.t. $x_1 < \dots < x_n$
 - ▶ $p_i > 0$ are *relative frequencies* i.e. $\sum_{i=1}^n p_i = 1$
 - ▶ n is the number of observations
- ▶ The space of all D is $\mathcal{D}$
 - ▶ The space of all D with $n \leq 2$ is $\ddot{\mathcal{D}}$
 - ▶ $\delta_x \in \mathcal{D}$ denotes the degenerate dataset $\{x; 1\}$
- ▶ The primitive is the **location mapping** $m : \mathcal{D} \rightarrow \mathbb{R}$
 - ▶ Coarser yet easier description of the data
 - ▶ Information loss is costly
 - ▶ The cost of representing x with y is $c(x, y)$

Setting (2)

- ▶ $c : \mathbb{R} \times \mathbb{R} \rightarrow \mathbb{R}^+$ is:
 1. continuous in each argument
 2. strictly convex in its second argument
 3. s.t. $c(x, y) = 0 \Leftrightarrow x = y$
 4. s.t. $c(x, y) = c(x + k, y + k)$ for any $k \in \mathbb{R}$
- ▶ The set of functions satisfying these conditions is $\mathcal{C}$
- ▶ On each dataset D , n different errors are committed
- ▶ The cost of representing D with $\{y; 1\}$ is:

$$C(D, y) := \sum_{i=1}^n p_i c(x_i, y)$$

- ▶ A loss function $c(x, y)$ **rationalizes** the map $m(D)$ if:

$$m(D) = \arg \min_y C(D, y)$$

Axioms

Axiom 1 (Certainty Matching): $m(\{x; 1\}) = x$

Axiom 2 (Continuity): For any $\{D_k\}_{k \in \mathbb{N}} \subseteq \ddot{\mathcal{D}}$ and $D \in \ddot{\mathcal{D}}$

$$D_k \rightarrow D \Rightarrow m(D_k) \rightarrow m(D)$$

Axiom 3 (Dominance): $\forall D, D' \in \ddot{\mathcal{D}}, D \succ_{FOSD} D' \rightarrow m(D) > m(D')$

Axiom 4 (Translation Invariance): $\forall k \in \mathbb{R}, D \in \ddot{\mathcal{D}}, m(D + k) = m(D) + k$

Axiom 5 (Odds Independence): For any $x_1 < x_2 < y < x_3 < x_4$, $\{x_2, x_3; p\}$,

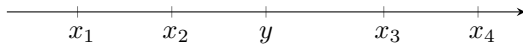
$\{x_2, x_4; q\}, \{x_1, x_3; r\}, \{x_1, x_4; s\}$ valued $y \Rightarrow$

$$\frac{q}{1-q} \cdot \frac{r}{1-r} = \frac{s}{1-s} \cdot \frac{p}{1-p}$$

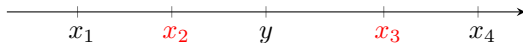
Axiom 6 (Reduction): $\forall D \in \mathcal{D}$ s.t. $m(D) = y$ and $n > 2$, $\exists D'$ s.t. $m(D') = y$ and $n' = n - 1$. The construction of D' follows one of two rules:

- Reduction by Cancellation
- Reduction by Aggregation

Odds Independence — Visualization

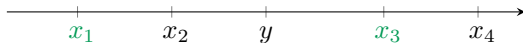


Odds Independence — Visualization



► $D = \{x_2, x_3; p, 1 - p\}$

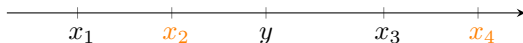
Odds Independence — Visualization



► $D = \{x_2, x_3; p, 1 - p\}$

► $D' = \{x_1, x_3; r, 1 - r\}$

Odds Independence — Visualization

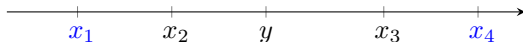


► $D = \{x_2, x_3; p, 1 - p\}$

► $D' = \{x_1, x_3; r, 1 - r\}$

► $D'' = \{x_2, x_4; q, 1 - q\}$

Odds Independence — Visualization



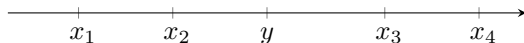
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Odds Independence — Visualization



► $D = \{x_2, x_3; p, 1 - p\}$

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$$m(D) = m(D') = m(D'') = m(D''') \Rightarrow \frac{\frac{p}{1-p}}{\frac{r}{1-r}} = \frac{\frac{q}{1-q}}{\frac{s}{1-s}}$$

Reduction by Cancellation

- If $y = x_k$ for some $x_k \in \text{supp}(D)$, then:

$$D' = \left\{ \{x_i\}_{i \neq k}; \left\{ \frac{p_i}{1 - p_k} \right\}_{i \neq k} \right\}$$

- If $D = \{1, 2, 3; \frac{1}{3}, \frac{1}{3}, \frac{1}{3}\}$ and $m(D) = 2$, then $D' = \{1, 3; \frac{1}{2}, \frac{1}{2}\}$

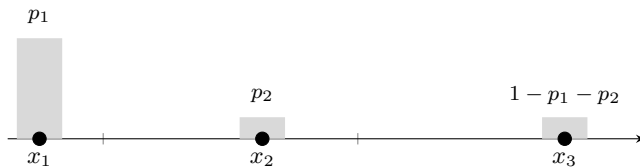
Reduction by Aggregation (1)

- Let $D = \{x_1, x_2, x_3; p_1, p_2\}$ with $m(D) = y \notin \text{supp}(D)$



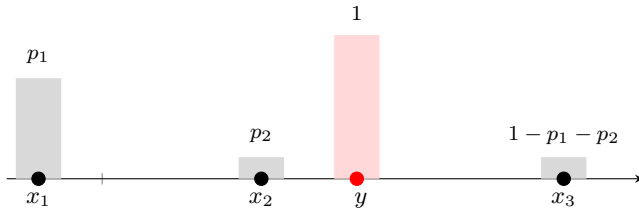
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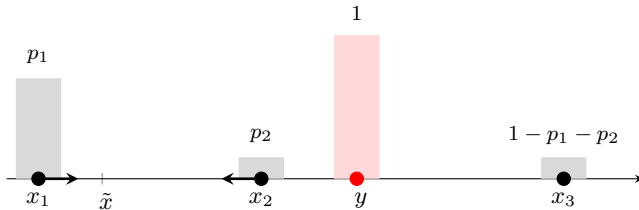
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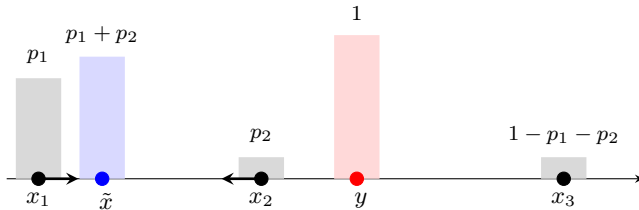
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Reduction by Aggregation (2)

If $y \notin \text{supp}(D)$, two adjacent values $(x_i, x_{i+1}) \in \text{supp}(D)$ **lying on the same side** of y are replaced by a unique value $\tilde{x}$ with frequency $(p_i + p_{i+1})$

To determine $\tilde{x}$:

- ▶ Fix an anchor k strictly on the opposite side of y from the pair
- ▶ for each x on the pair's side, let $s_y(x; k) \in (0, 1)$ be the (unique) weight on x such that $m\{s_y(x; k)\delta_x + [1 - s_y(x; k)]\delta_k\} = y$
- ▶ Then, $\tilde{x}$ is the unique value satisfying:

$$\frac{1 - s_y(\tilde{x}; k)}{s_y(\tilde{x}; k)} = \frac{p_i}{p_i + p_{i+1}} \cdot \frac{1 - s_y(x_i; k)}{s_y(x_i; k)} + \frac{p_{i+1}}{p_i + p_{i+1}} \cdot \frac{1 - s_y(x_{i+1}; k)}{s_y(x_{i+1}; k)} \quad (1)$$

Reduction by Aggregation (3)

- ▶ If $m(D) \notin \text{supp}(D)$, Reduction is a *similarity-preserving* principle
- ▶ $s_y(x; k)$ measures the similarity of x and y : the more weight x must carry to drag a two-point location index onto y , the closer x and y are
- ▶ The corresponding odds $\frac{1-s_y(x; k)}{s_y(x; k)}$ are a dissimilarity index
- ▶ Equation (1) requires the similarity of $\tilde{x}$ and y to be the weighted average of the one of x_i and y and x_{i+1} and y , respectively

Representation Theorem

Theorem

The following are equivalent:

1. *m satisfies Axioms 1-6*
2. *$\exists c \in \mathcal{C}$ such that:*

$$m(D) = \arg \min_y \sum_{i=1}^n p_i c(x_i, y) \quad \forall D \in \mathcal{D}$$

Also, c is unique up to any positive multiplicative constant

- ▶ Axiomatic foundation of translation-invariant M -estimators (Huber, 1964)
- ▶ $m(D)$ is the optimal forecast for a given scoring function $c(x, y)$
- ▶ Asymmetric version of Fishburn (1986) (not nested)

Proof Sketch — Sufficiency (1)

1. Define frequencies:
- ▶ $\forall x^- < 0$, find p^- s.t. $m(\{x^-, 1, p^-\}) = 0$
 - ▶ $\forall x^+ > 0$, find p^+ s.t. $m(\{\hat{x}^-, x^+, p^+\}) = 0$ for a $\hat{x}^-$ (irrelevant by *Odds Independence*)
 - ▶ *Continuity* and *Dominance* guarantee existence and uniqueness

2. Constructing c' around 0:
- ▶ Normalize $c'(1, 0) = -1$
 - ▶ Using FOC: $c'_l(x, 0) = \frac{1-p^-}{p^-}$
 - ▶ Using FOC: $c'_r(x, 0) = -\frac{p^+}{1-p^+} \cdot \frac{1-\hat{p}^-}{\hat{p}^-}$

3. Generalize to any center:
- ▶ Use *Translation Invariance* to shift the centre from 0 to any value y

Proof Sketch — Sufficiency (2)

4. Prove c' is well-behaved:
 - ▶ We show that c' is:
 - ▶ continuous in each argument
 - ▶ strictly decreasing in x
 - ▶ strictly increasing in y

5. Get total loss function:
 - ▶ Integrate $c'(x, y)$ w.r.t. y to get $c(x, y)$
 - ▶ Prove that $c \in \mathcal{C}$
 - ▶ In particular, *Certainty Matching* ensures $c(x, y) \geq 0$, with equality iff $x = y$

6. Induction step:
 - ▶ If c rationalizes m for a dataset with n entries, *Reduction* guarantees it also rationalizes m for a dataset with $n + 1$ entries

Conclusions

- ▶ I provide an axiomatic characterization of location indices as minimizers of continuous, (strictly) convex loss functions
- ▶ Why this matters:
 1. Provides a unified framework connecting decision theory, statistics, and machine learning
 2. Establishes that the choice of location index is fundamentally a choice about how to penalize prediction errors
 3. Grounds the behavioral study of risk preferences in the geometrical features of loss functions
- ▶ Anomalies in decision making under risk may reflect departures from the implicit loss function assumed by classical EU theory

Related Literature

- ▶ *Means*: Kolmogorov (1930), Nagumo (1930), Nagumo (1931), De Finetti (1931), Fishburn (1986), Hong (1983), Muliere and Parmigiani (1993), Chambers (2009), De Carvalho (2016), Singpurwalla and Lai (2020), Bellini and Peri (2022)
- ▶ *Robust Estimators*: Huber (1964), Breckling and Chambers (1988), van der Vaart and Wellner (1996), Chen (1996), Newey and Powell (1987), Ziegel (2016)
- ▶ *Quantiles and Expectiles*: Koenker and Bassett Jr (1978), Pennecchi and Callegaro (2006), Rostek (2010), Ehm et al. (2016), Bellini and Di Bernardino (2017), Daouia et al. (2020), Philipps (2022), Bellini et al. (2025)
- ▶ *Certainty Equivalents*: Huygens (1657), Bernoulli (1738), ?, von Neumann and Morgenstern (1944), de Finetti and Emanuelli (1967), Kahneman et al. (1979) Tversky and Kahneman (1992), Harbaugh et al. (2010), Oprea (2024), Roorda and Joosten (2025)

Thank you!

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